

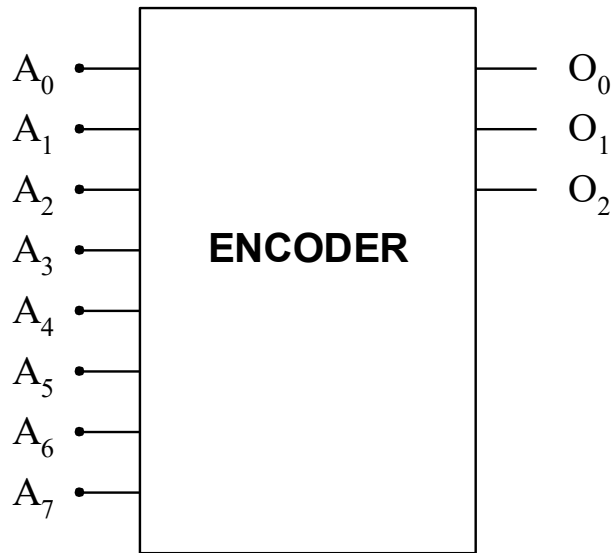
Decoders and Encoders

Encoder/Decoder

ENCODER- a digital circuit that produces a binary output code depending on which of its inputs are activated.

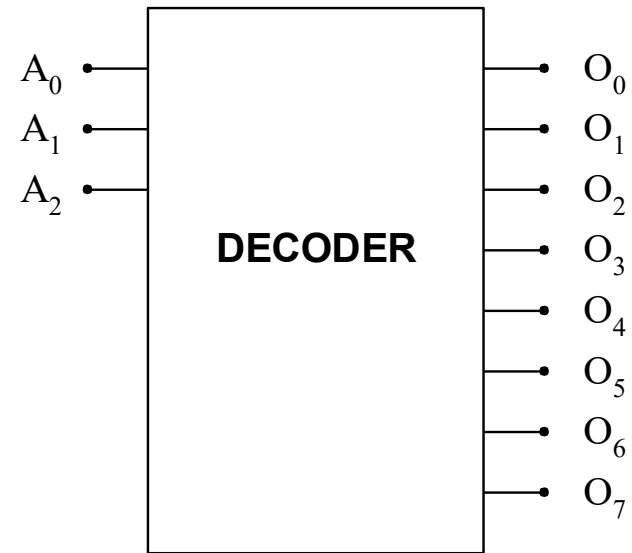
DECODER- a digital circuit that converts an input binary code into a single numeric output.

ENCODERS AND DECODERS



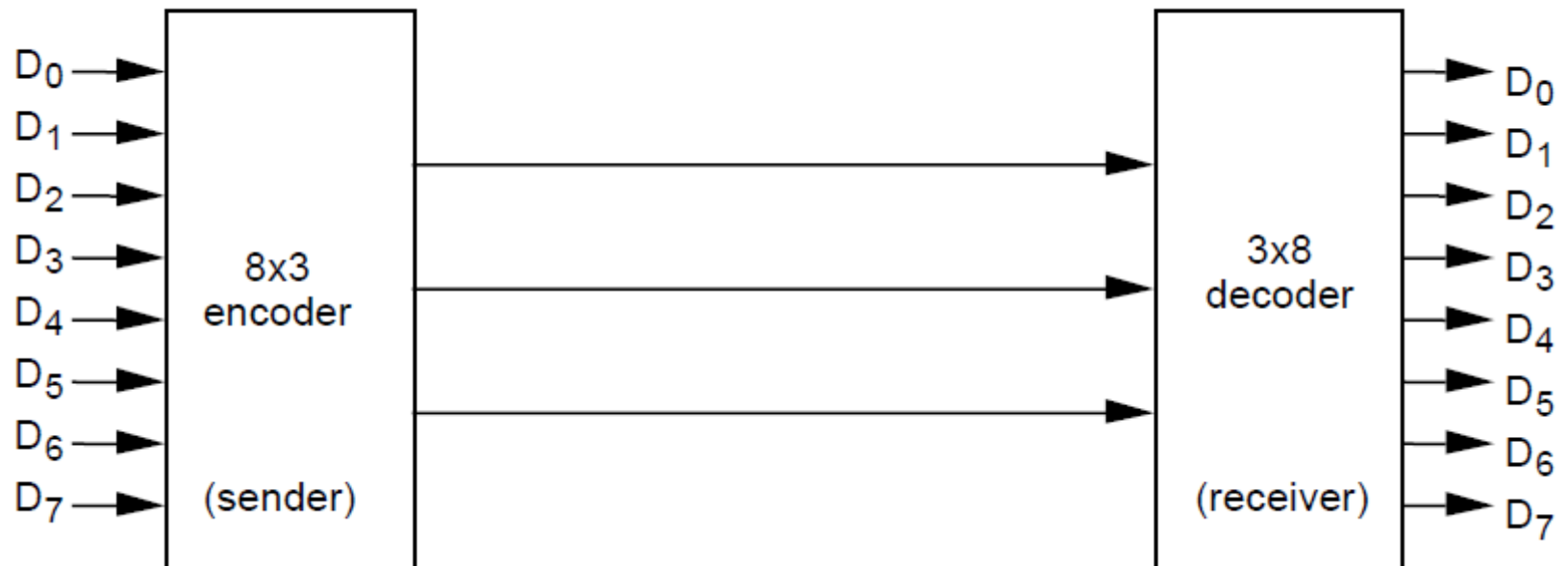
ONLY ONE INPUT
ACTIVATED AT A TIME

BINARY CODE OUTPUT



BINARY CODE INPUT

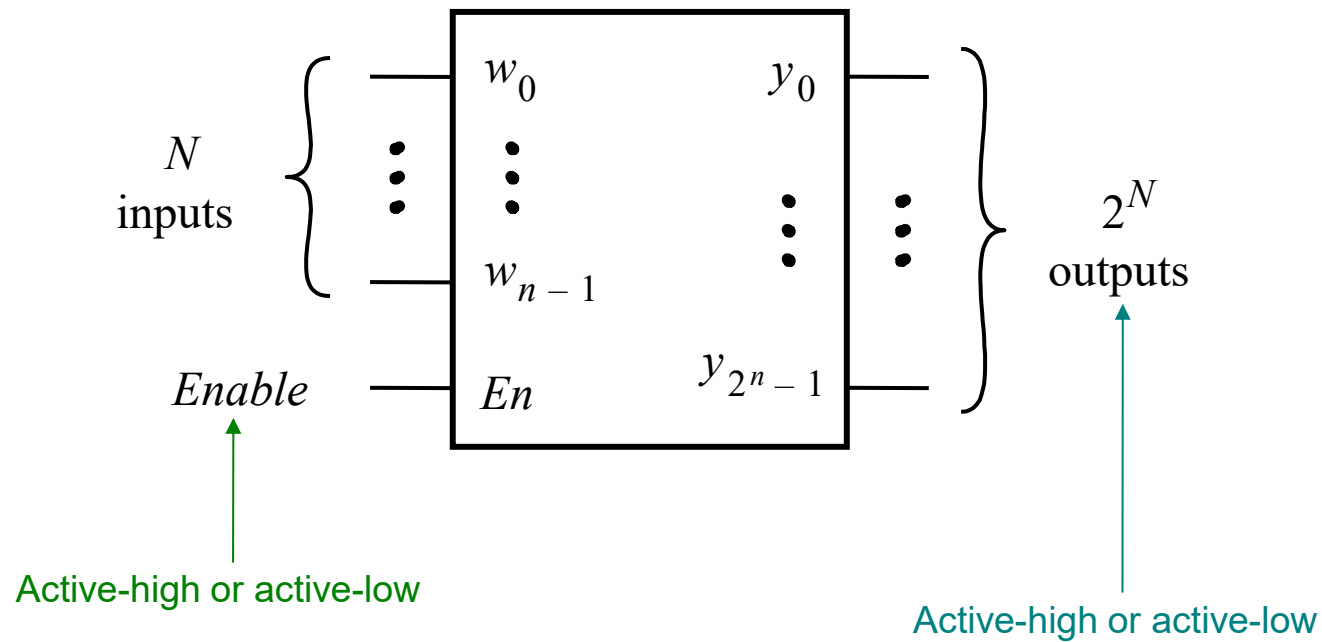
ONLY ONE OUTPUT
ACTIVATED AT A TIME



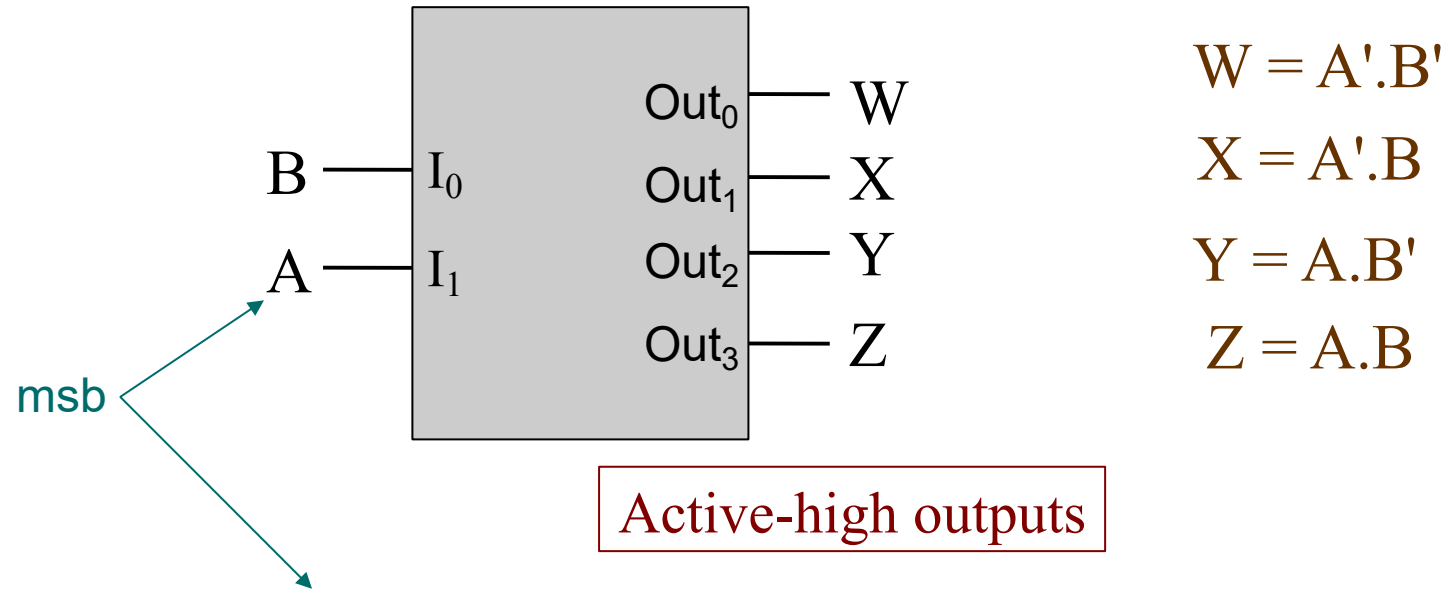
Main function of encoder and decoder

The purpose is to reduce the number of wires required for interconnection.

Decoder: N-to- 2^N



Decoders

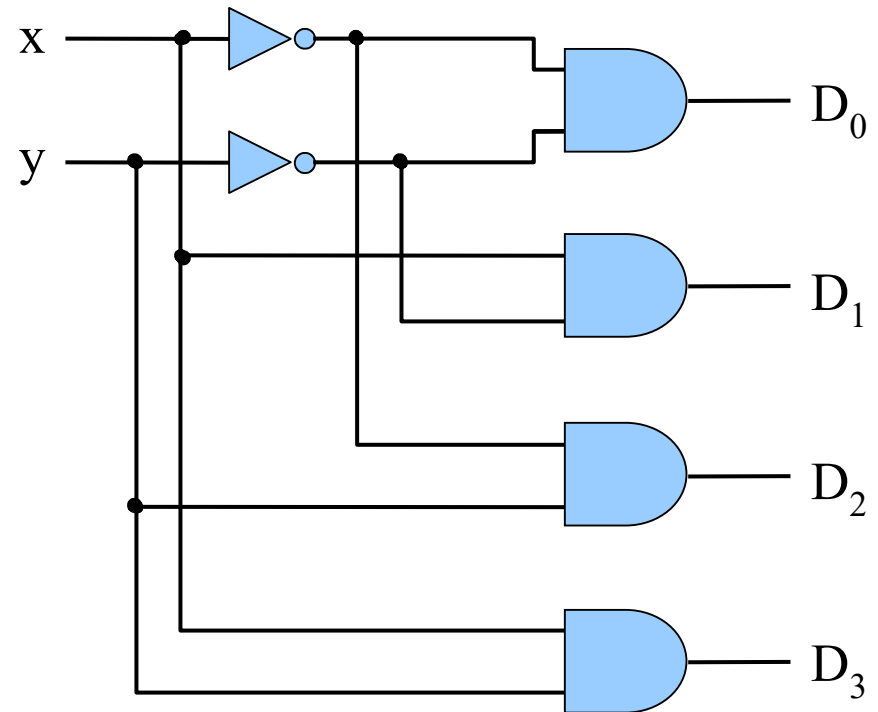


A	B	W	X	Y	Z
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Decoder: 2-to-4

y	x	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

active-high output



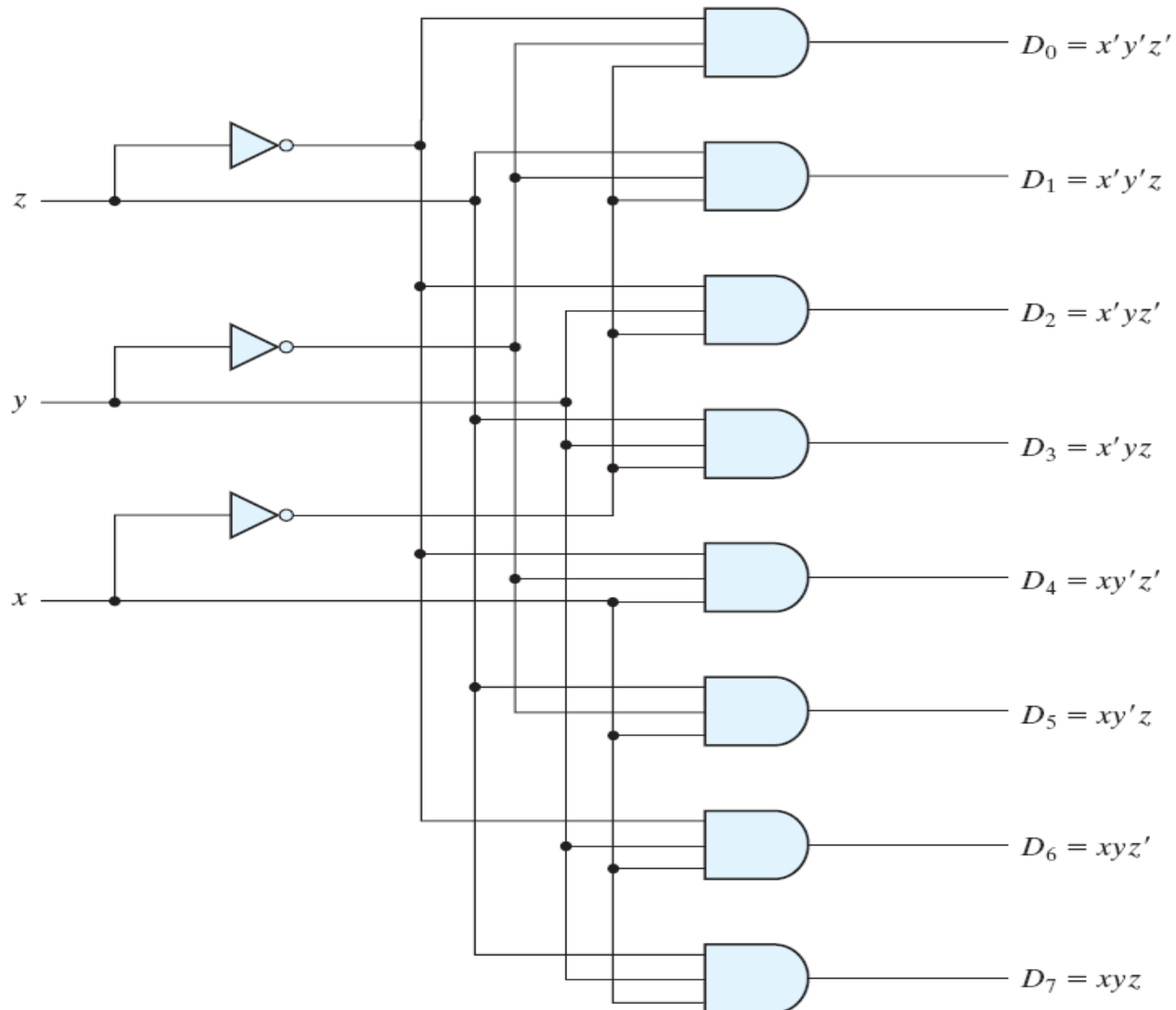
Decoder: 3-to-8

Table 4.6

Truth Table of a Three-to-Eight-Line Decoder

Inputs			Outputs							
<i>x</i>	<i>y</i>	<i>z</i>	<i>D</i> ₀	<i>D</i> ₁	<i>D</i> ₂	<i>D</i> ₃	<i>D</i> ₄	<i>D</i> ₅	<i>D</i> ₆	<i>D</i> ₇
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

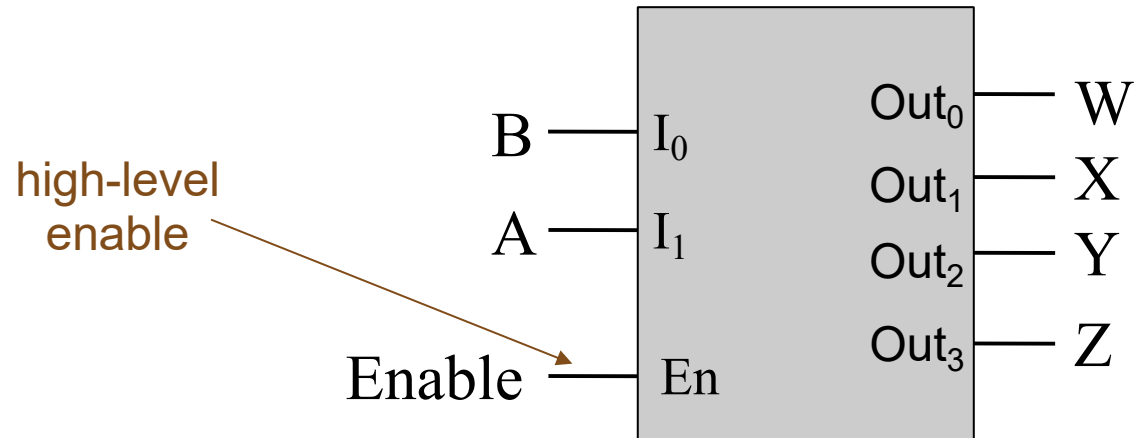
Decoder: 3-to-8



Definitions

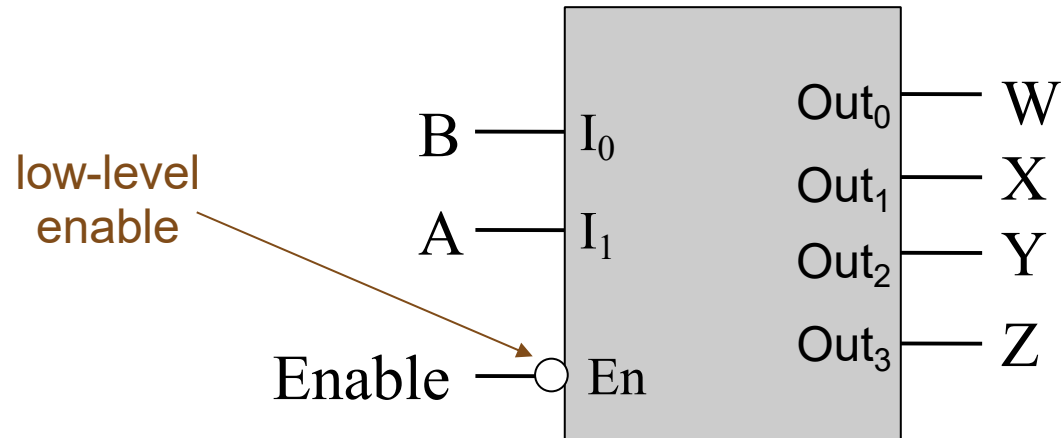
- Active – the function is being performed
 - Active low – active when the output is 0.
 - Active high – active when the output is 1.
- Enabled – active circuit can perform its function
 - Low-level enabled
 - circuit performs its function when Enable = 0.
 - High-level enabled
 - circuit performs its function when Enable = 1.

Decoder with Enable



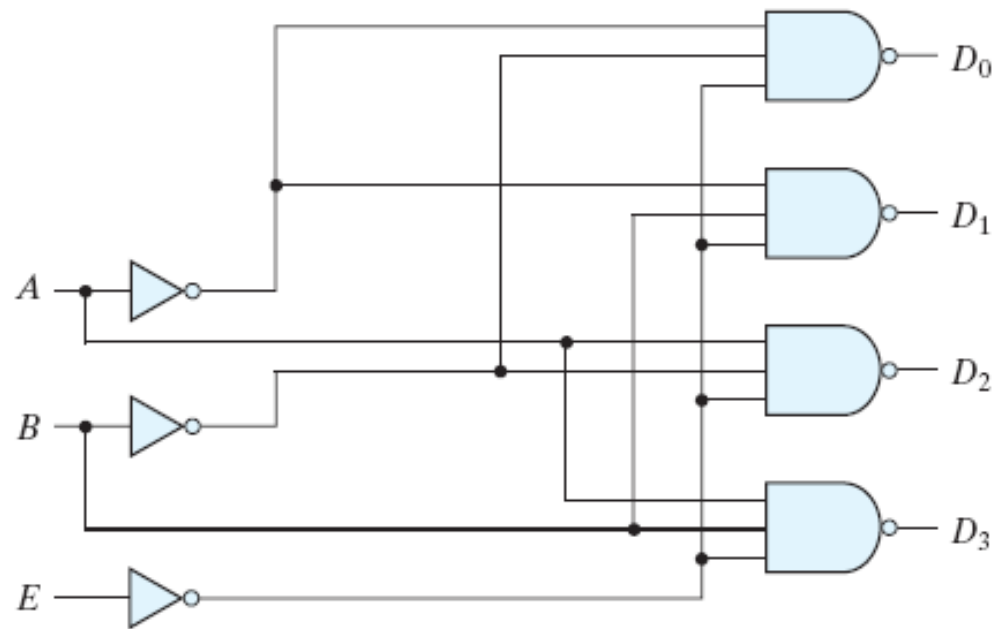
	En	A	B	W	X	Y	Z
enabled	1	0	0	1	0	0	0
	1	0	1	0	1	0	0
	1	1	0	0	0	1	0
	1	1	1	0	0	0	1
disabled	0	x	x	0	0	0	0

Decoder with Enable



	En	A	B	W	X	Y	Z
enabled	0	0	0	1	0	0	0
	0	0	1	0	1	0	0
	0	1	0	0	0	1	0
	0	1	1	0	0	0	1
disabled	1	x	x	0	0	0	0

Decoder: 2-to-4 with Enable



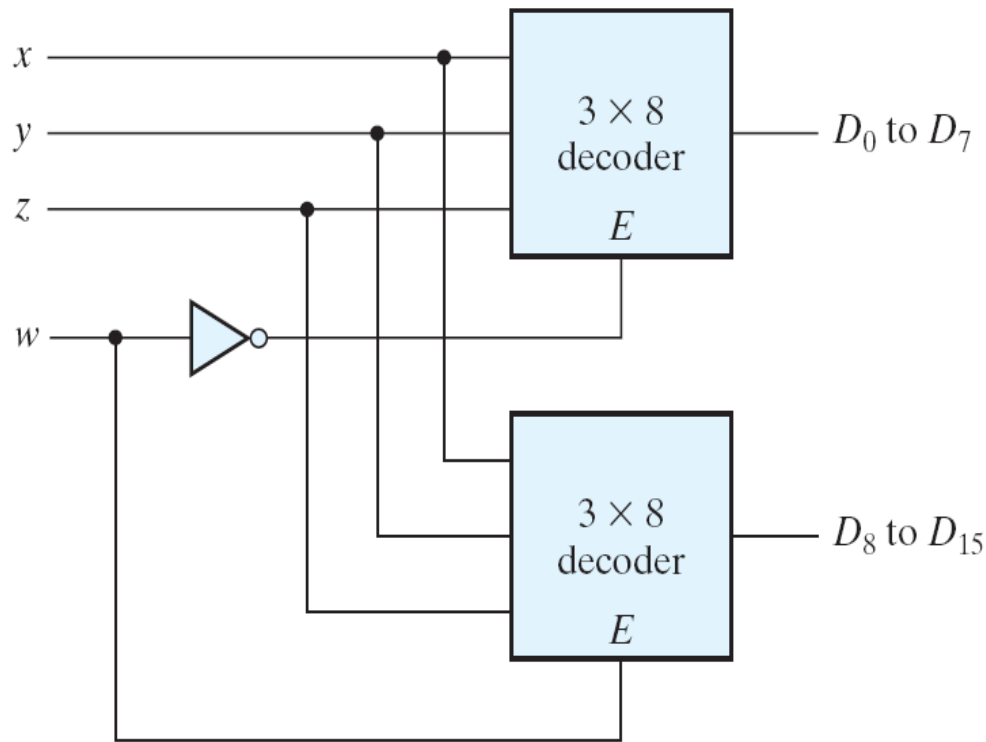
(a) Logic diagram

E	A	B	D_0	D_1	D_2	D_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0

(b) Truth table

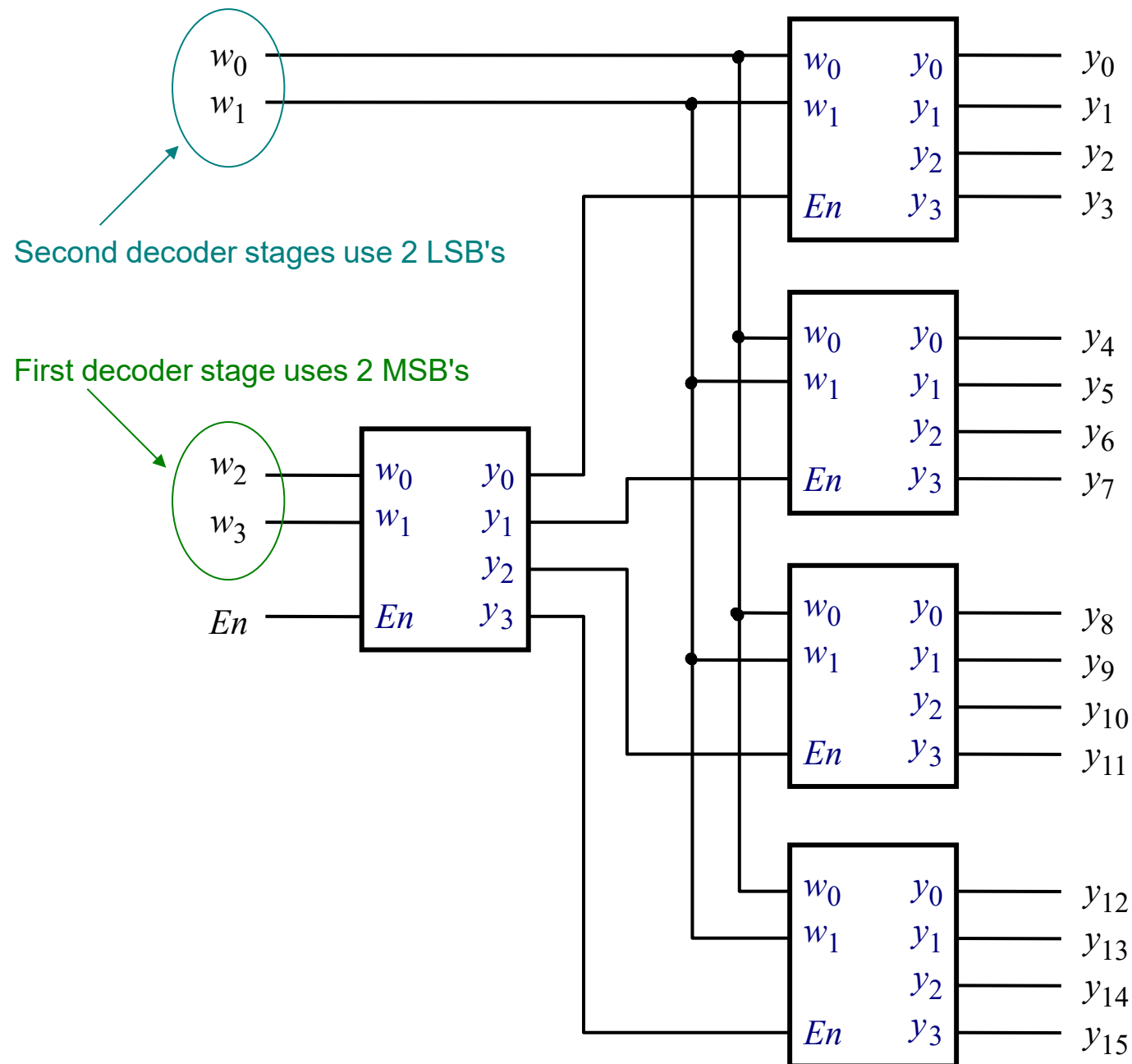
active-low output

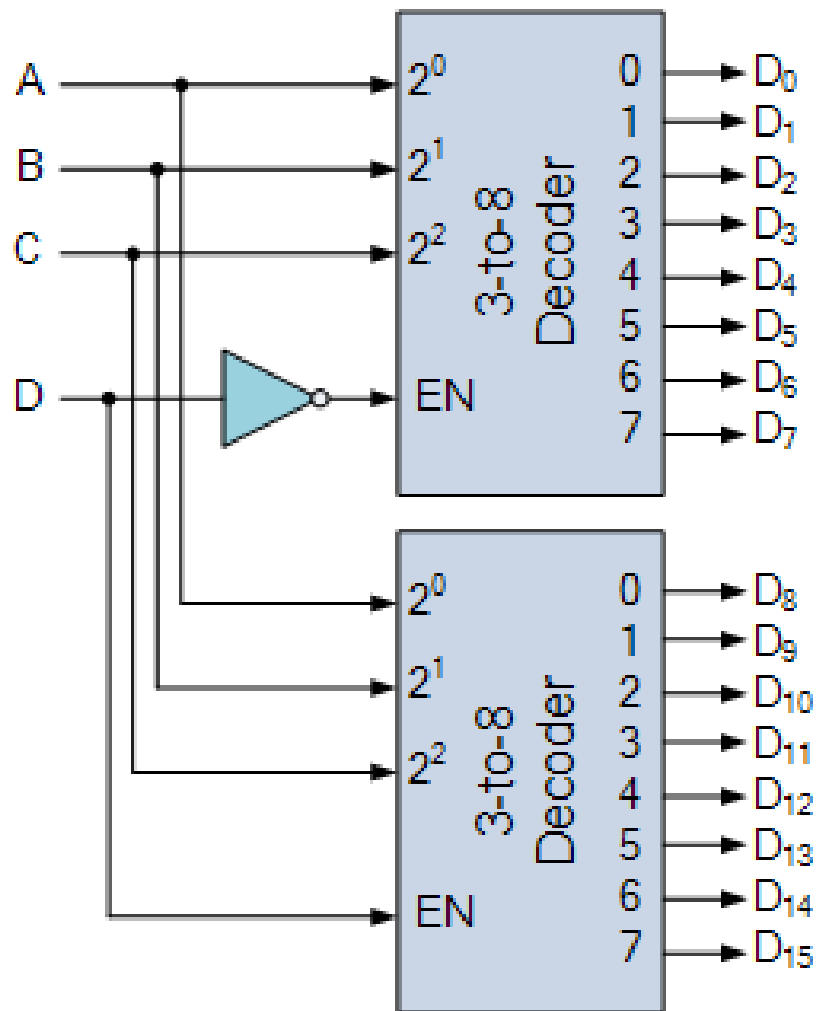
4*16 Decoder using 3* 8 Decoder



W	X	Y	Z	Output
0	0	0	0	D0
0	0	0	1	D1
0	0	1	0	D2
0	0	1	1	D3
0	1	0	0	D4
0	1	0	1	D5
0	1	1	0	D6
0	1	1	1	D7
1	0	0	0	D8
1	0	0	1	D9
1	0	1	0	D10
1	0	1	1	D11
1	1	0	0	D12
1	1	0	1	D13
1	1	1	0	D14
1	1	1	1	D15

4*16 Decoder using 2* 4 Decoder





4-to-16 Line Decoder Implemented
with two 3-to-8 Decoders

D	C	B	A	Output
0	0	0	0	D0
0	0	0	1	D1
0	0	1	0	D2
0	0	1	1	D3
0	1	0	0	D4
0	1	0	1	D5
0	1	1	0	D6
0	1	1	1	D7
1	0	0	0	D8
1	0	0	1	D9
1	0	1	0	D10
1	0	1	1	D11
1	1	0	0	D12
1	1	0	1	D13
1	1	1	0	D14
1	1	1	1	D15

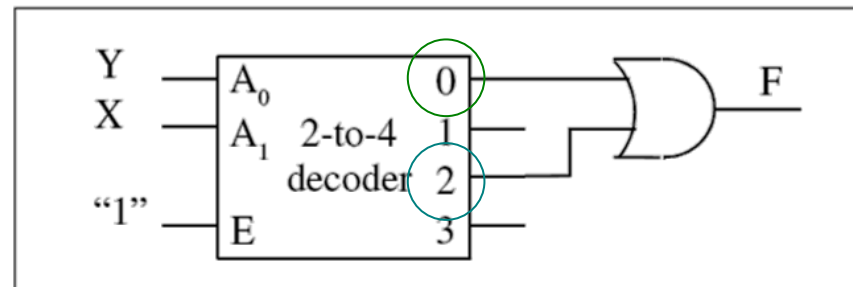
Decoders

Designing Logic Circuits
using Decoders

Decoders: Designing Logic Circuits

$$F = \Sigma m(0,2)$$

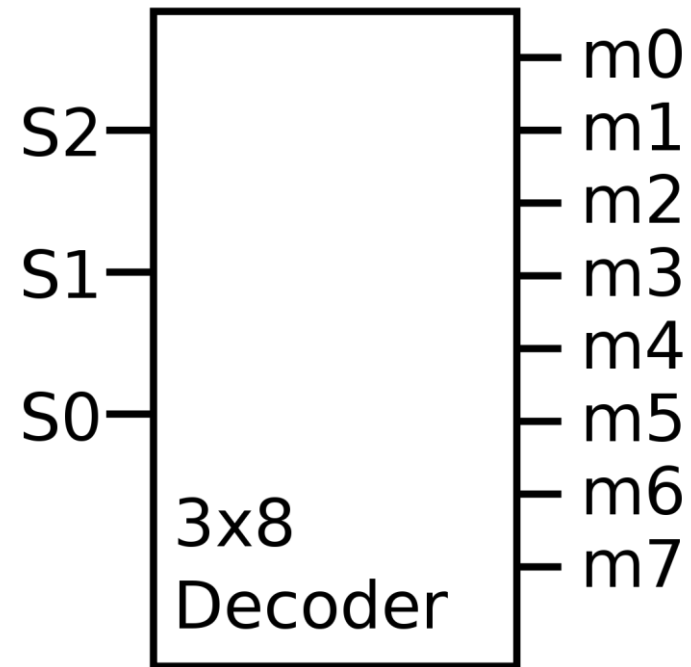
A	B	F	O ₃	O ₂	O ₁	O ₀
0	0	1				1
0	1	0			1	
1	0	1		1		
1	1	0	1			



Design all logic gates using a Decoder.

Design a circuit, using a 3-to-8 Decoder to realize the Boolean expression given below.

$$F_{X,Y,Z} = \sum m(1,2,5,7)$$

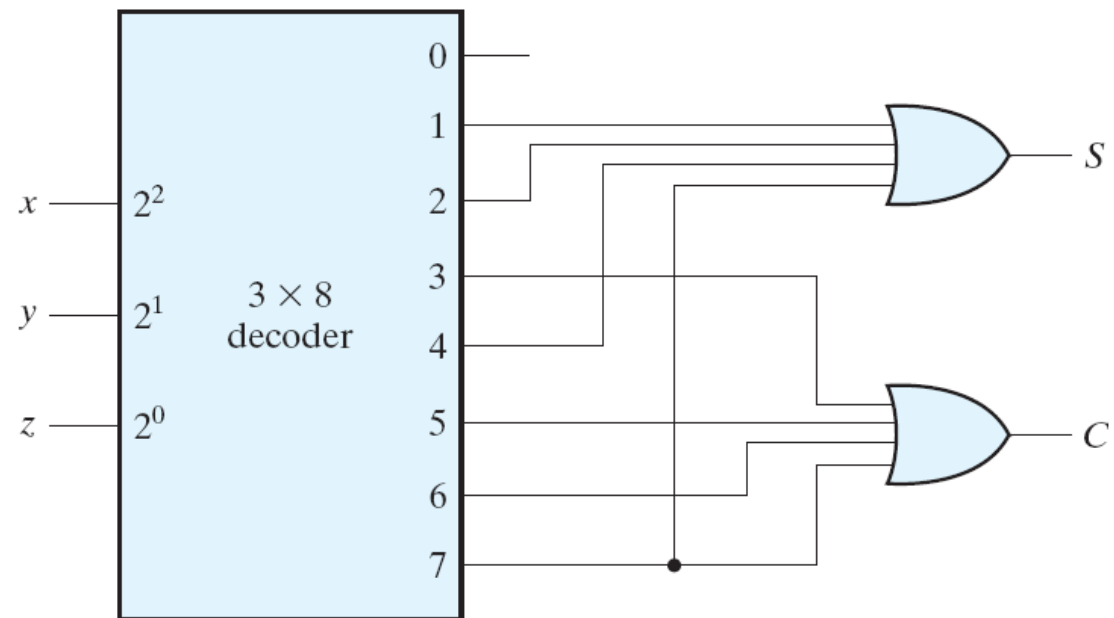


Decoders: Designing Logic Circuits

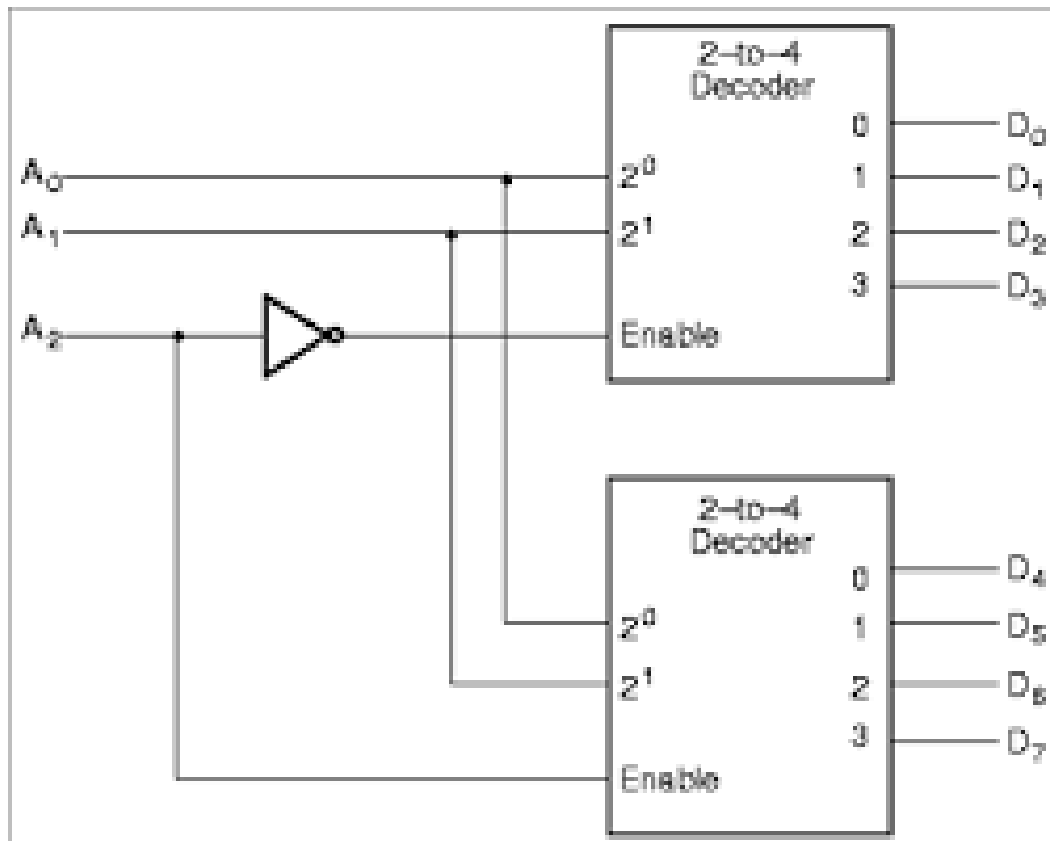
Exercise:

Design a Half Adder using a Decoder

Design a Full Adder using a 3-to-8 Decoder.



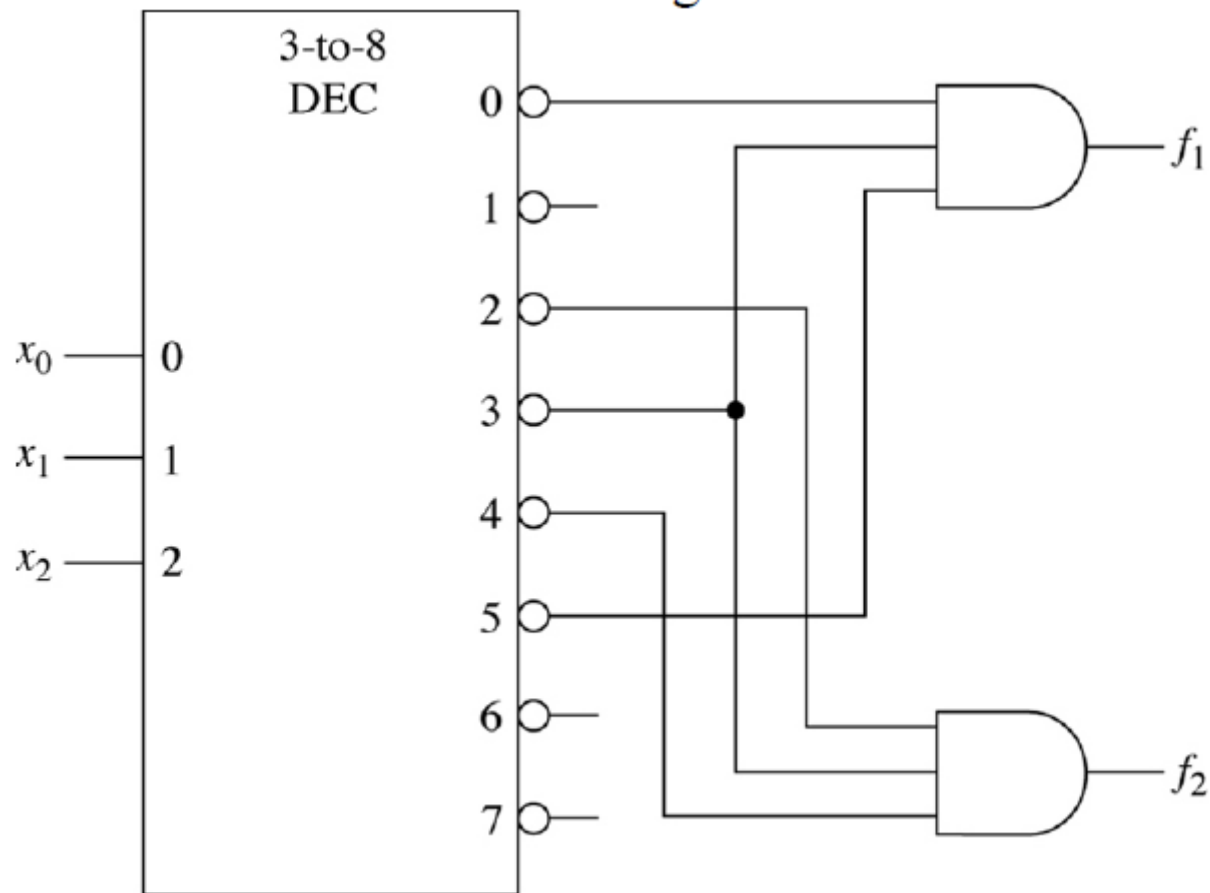
Design a Full Adder using a 2-to-4 Decoder.



Add two OR gate to design sum and carry equation.

Realization of the pair of maxterm canonical expressions
 $f_1(x_2, x_1, x_0) = \Pi M(0, 3, 5)$ and $f_2(x_2, x_1, x_0) = \Pi M(2, 3, 4)$ with a
3-to-8-line decoder and two and-gates.

Realization of the pair of maxterm canonical expressions $f_1(x_2, x_1, x_0) = \prod M(0, 3, 5)$ and $f_2(x_2, x_1, x_0) = \prod M(2, 3, 4)$ with a 3-to-8-line decoder and two and-gates.



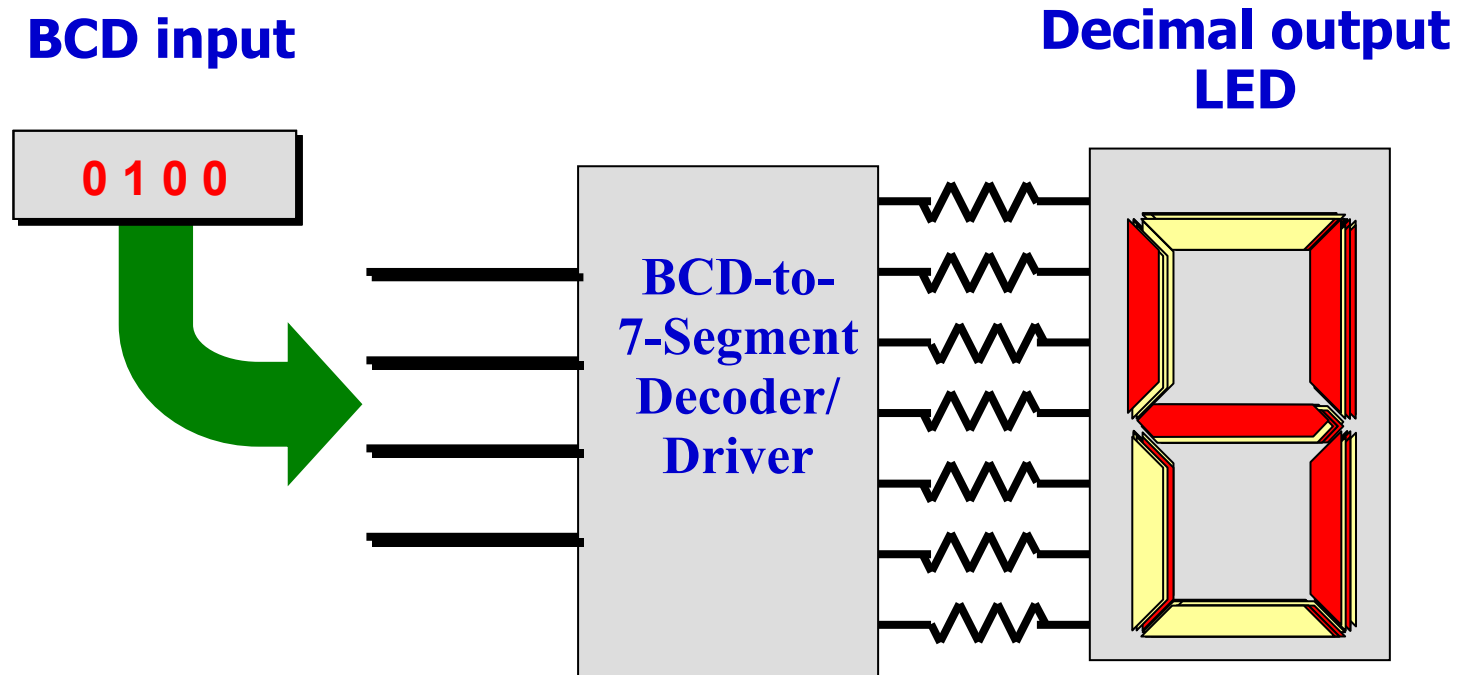
Decoders: Designing Logic Circuits

Design a circuit, using 2-to-4 Decoders to realize the Boolean expression given below.

$$F_{X,Y,Z} = \Sigma m(1,2,5,7)$$

Use two 2*4 decoder with enable Z

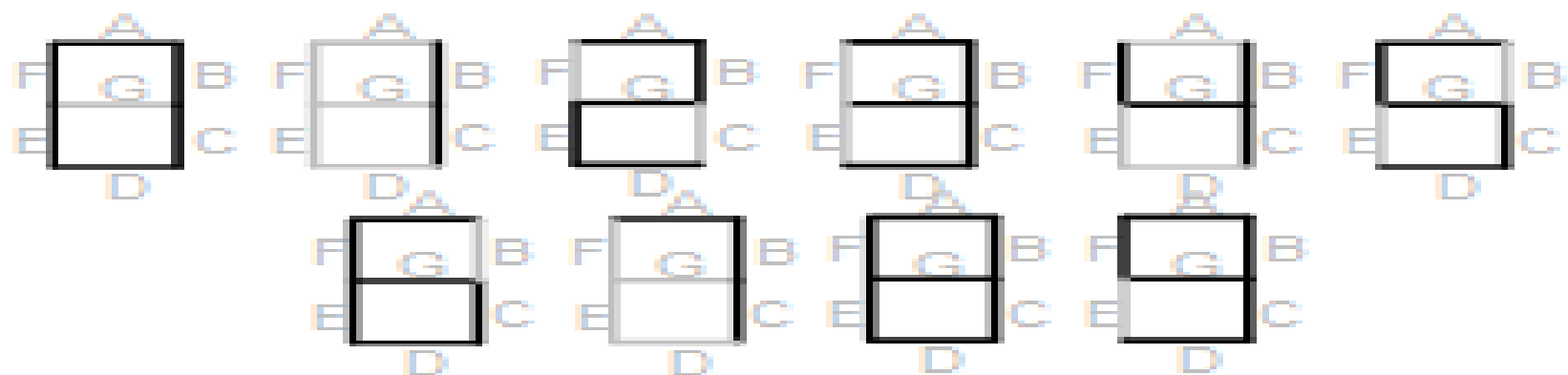
DECODERS: BCD TO 7-SEGMENT DECODER/DRIVER



- **Electronic decoders are available in IC form.**
- **This decoder translates from BCD to decimal.**
- **Decimals are shown on 7-segment LED display.**
- **This IC also drives the 7-segment LED display.**

Binary Inputs				Decoder Outputs							7-Segment Display Outputs
D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	1	1	1	1	1	1	0	0
0	0	0	1	0	1	1	0	0	0	0	1
0	0	1	0	1	1	0	1	1	0	1	2
0	0	1	1	1	1	1	1	0	0	1	3
0	1	0	0	0	1	1	0	0	1	1	4
0	1	0	1	1	0	1	1	0	1	1	5
0	1	1	0	1	0	1	1	1	1	1	6
0	1	1	1	1	1	1	0	0	0	0	7
1	0	0	0	1	1	1	1	1	1	1	8
1	0	0	1	1	1	1	1	0	1	1	9

	BCD	A	B	C	D	E	F	G
0	0000	1	1	1	1	1	1	
1	0001		1	1				
2	0010	1	1		1	1		1
3	0011	1	1	1	1			1
4	0100		1	1			1	1
5	0101	1		1	1		1	1
6	0110	1		1	1	1	1	1
7	0111	1	1	1				
8	1000	1	1	1	1	1	1	1
9	1001	1	1	1	1		1	1
10	1010	x	x	x	x	x	x	x
11	1011	x	x	x	x	x	x	x
12	1100	x	x	x	x	x	x	x
13	1101	x	x	x	x	x	x	x
14	1110	x	x	x	x	x	x	x
15	1111	x	x	x	x	x	x	x

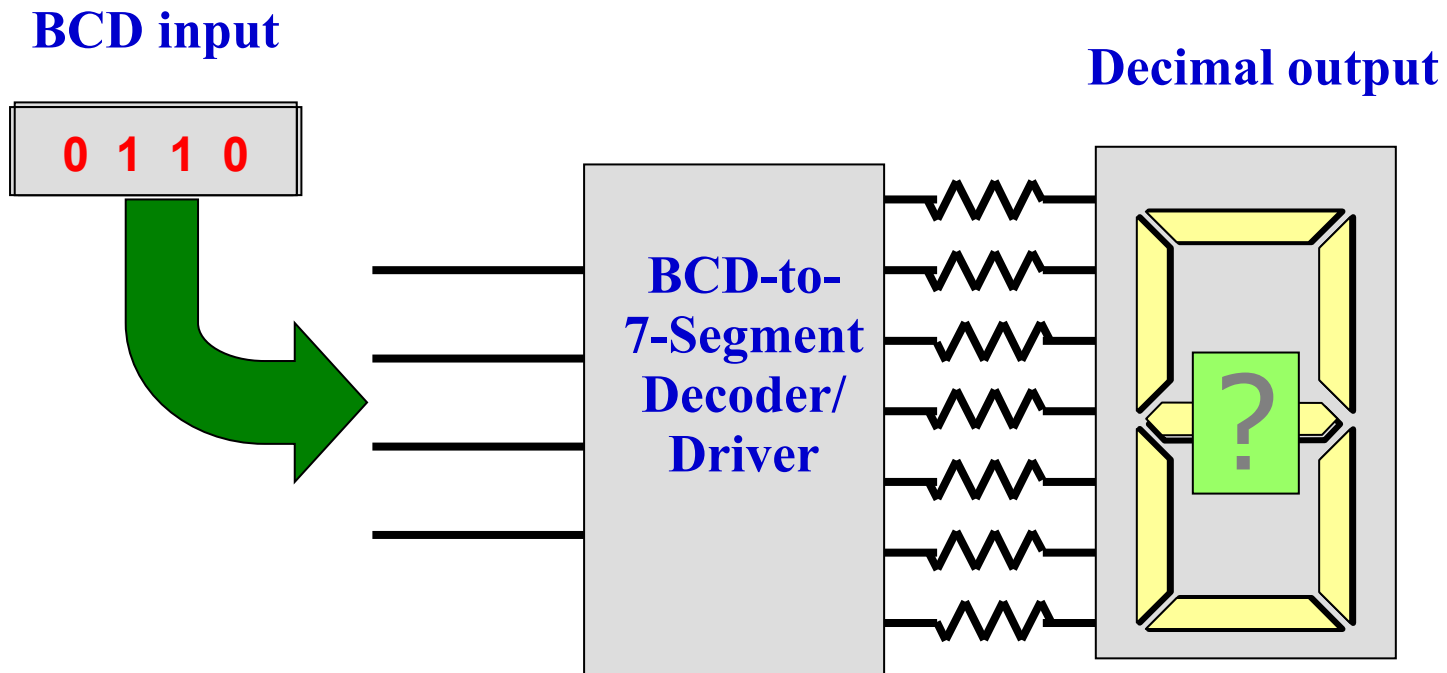




TEST

Q #5- What is the decimal output from the decoder that appears on the 7-segment display?

Answer: 6



Encoders

Encoder

An encoder outputs the binary value of the active input.

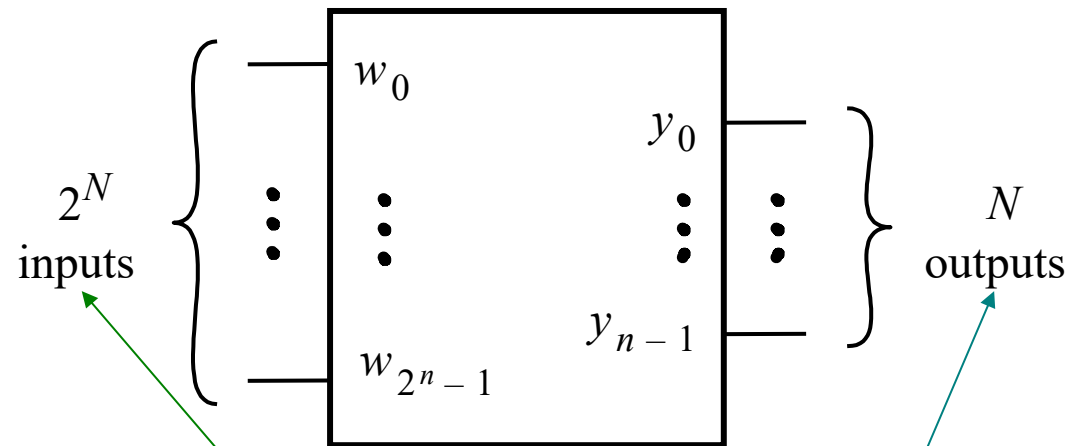
An encoder has 2^N inputs and N outputs.

If we record any music in any recorder, it means that we are giving data to a recorder. So, such process is called encoding.

Encoder

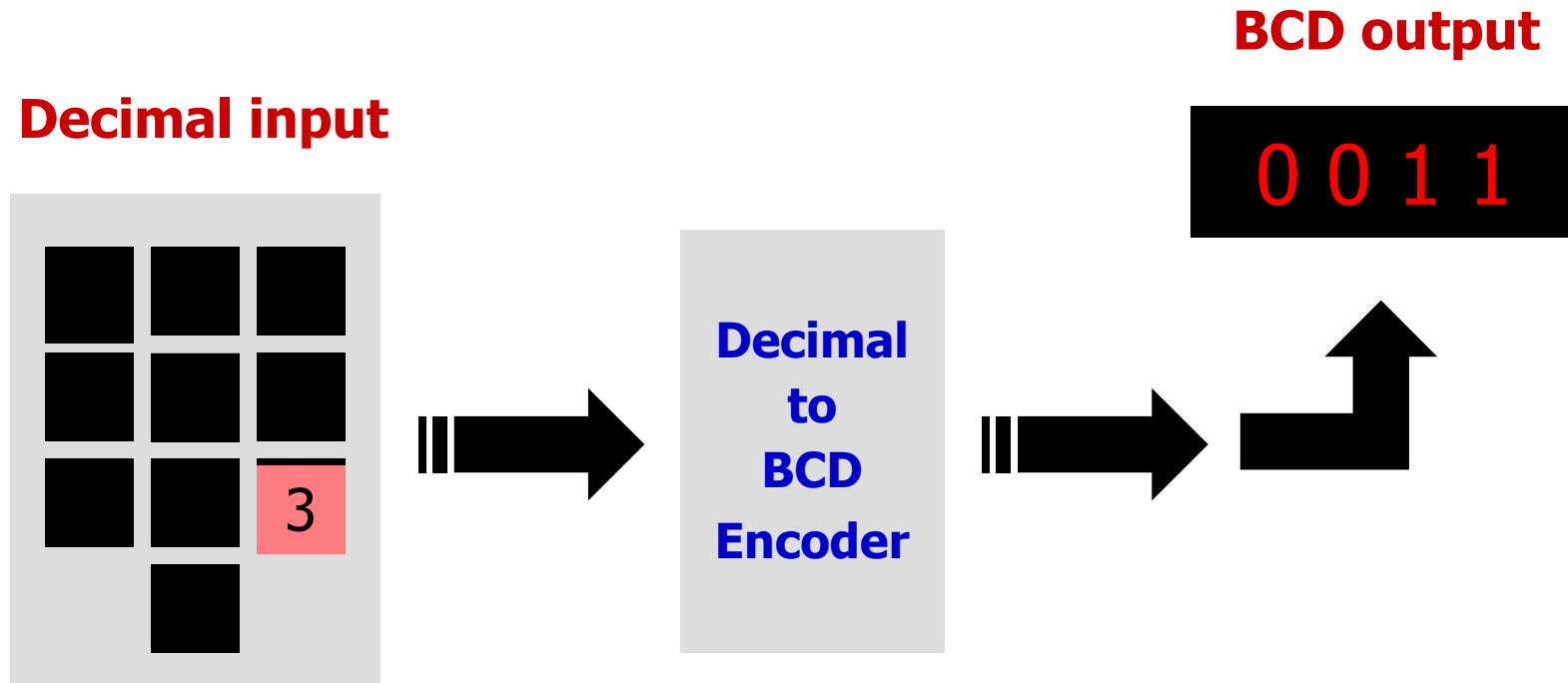
- Inverse operation of the decoder
 - Decoder selects one output based on the input binary number.
 - Encoder outputs a binary number based on the selected input.
- Issues
 - What if more than one input is active?
 - What if no inputs are active?

Encoder: 2^N -to- N



The active (or selected) input is encoded on the n outputs.

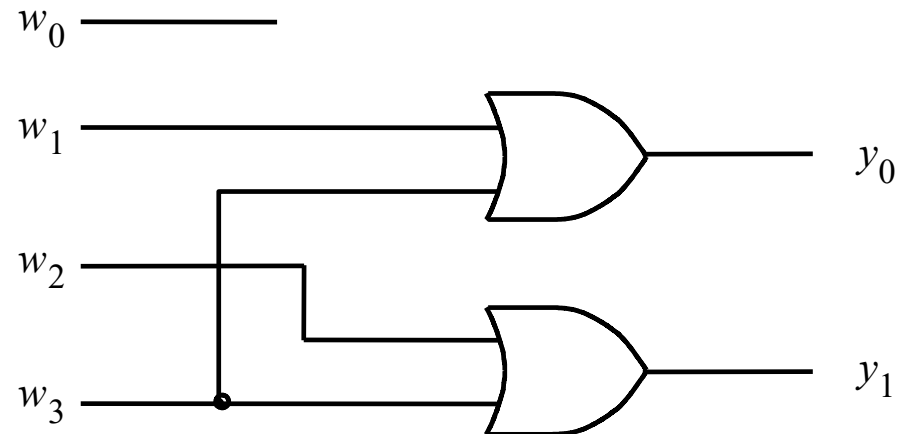
ELECTRONIC ENCODER - DECIMAL TO BCD



- Encoders are available in IC form.
- This encoder translates from decimal input to BCD output.

Encoder: 4-to-2

w_3	w_2	w_1	w_0	y_1	y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1



Encoder: 8-to-3

Table 4.7

Truth Table of an Octal-to-Binary Encoder

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

Priority Encoder

- Higher-order input has priority of lower-order input
 - Order of the input determined by its binary value
 - I0 has binary value 000 (0)
 - I1 has binary value 001 (1)
 - I2 has binary value 010 (2)
 - I3 has binary value 011 (3) etc.
 - So,
 - I1 selected over I0
 - I2 selected over I1 and I0
 - I3 selected over I2, I1, and I0
 - etc.

Priority Encoders

- If more than one input is active, the higher-order input has priority over the lower-order input.
 - The higher value is encoded on the output
- A valid indicator, d , is included to indicate whether or not the output is valid.
 - Output is invalid when no inputs are active
 - $d = 0$
 - Output is valid when at least one input is active
 - $d = 1$

Why is the valid indicator needed?

Priority Encoder

- Valid indicator
 - Output signal of the (priority) encoder that indicates the validity of the encoded output
 - Encoded output is invalid when no inputs are selected or when the encoder is disabled
 - $V = 0$ (indicates invalid encoded output; active high)
 - Encoded output when one, or more, input(s) is (are) selected, and encoder is enabled
 - $V = 1$ (indicates valid encoded output; active high)

4-to-2 Priority Encoder

Table 4.8

Truth Table of a Priority Encoder

Inputs				Outputs		
D_0	D_1	D_2	D_3	x	y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

Valid indicator

invalid

valid

4-to-2 Priority Encoder

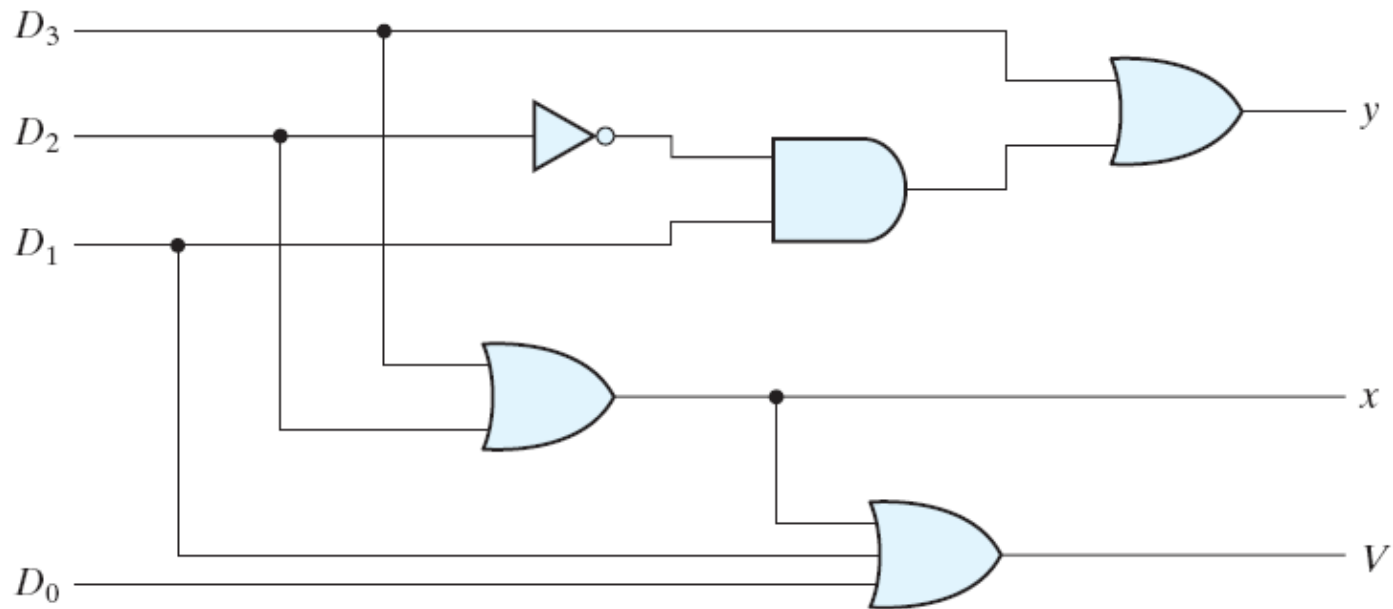
D_0D_1		D_2D_3			
		00	01	11	10
D_0	00	m_0 X	m_1 1	m_3 1	m_2 1
	01	m_4	m_5 1	m_7 1	m_6 1
	11	m_{12}	m_{13} 1	m_{15} 1	m_{14} 1
	10	m_8	m_9 1	m_{11} 1	m_{10} 1
		D_3			

$x = D_2 + D_3$

D_0D_1		D_2D_3			
		00	01	11	10
D_0	00	m_0 X	m_1 1	m_3 1	m_2
	01	m_4 1	m_5 1	m_7 1	m_6
	11	m_{12} 1	m_{13} 1	m_{15} 1	m_{14}
	10	m_8	m_9 1	m_{11} 1	m_{10}
		D_3			

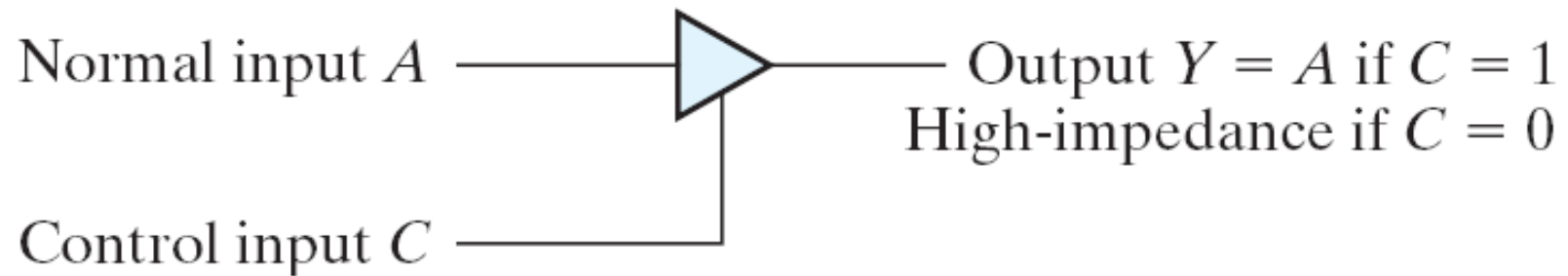
$y = D_3 + D_1D'_2$

4-to-2 Priority Encoder

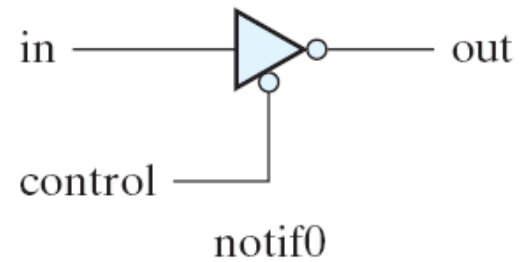
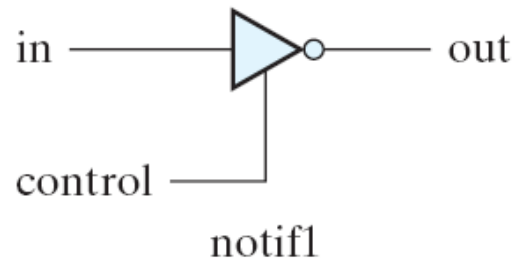
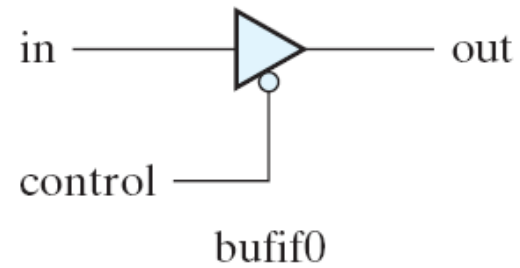
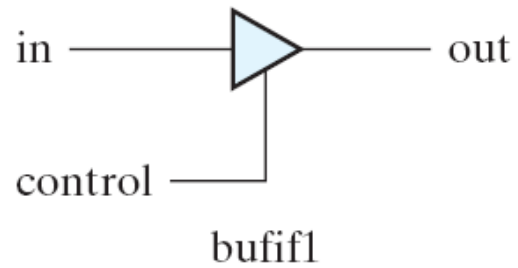


Tri-state Devices

Tri-state Devices



Tri-state Devices

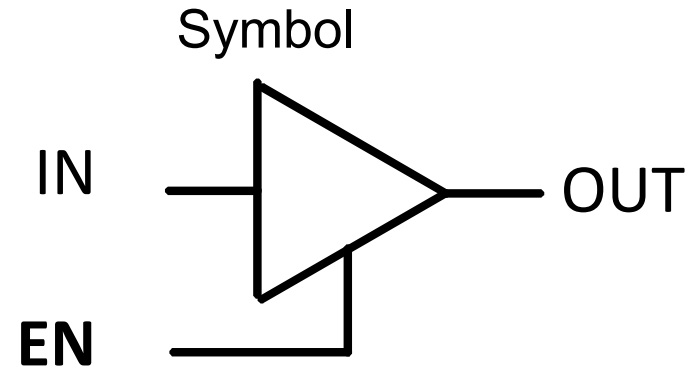


Hi-Impedance Output

- Logic gates introduced thus far ...
 - Have 1 and 0 output values
 - **Cannot** have their outputs connected together
- Three-state logic adds a third logic value:
 - Hi-Impedance output: **Z**
- What is Hi-Impedance output?
 - The output appears to be **disconnected** from the input
 - Behaves as an **open circuit** between gate input & output
- Hi-Z state makes a gate output behave differently:
 - Three output values: 1, 0, and Z
 - Hi-impedance gates **can connect** their outputs together

The 3-State Buffer

- IN = data input
- EN = Enable control input
- OUT = data output
- If EN = 0 then OUT = HI-Z
 - Regardless of the value on IN
 - Output disconnected from input
- If EN = 1, then OUT = IN
 - Output follows the input value
- Variations:
 - EN can be inverted
 - OUT can be inverted
 - By addition of bubbles to signals



Truth Table

EN	IN	OUT
0	X	Z
1	0	0
1	1	1